BLOCKCHAIN STARTUP CONCEPT

CertiChain: Decentralized Verification of Educational Documents

Secure diploma verification through blockchain hashing, without disclosing personal data off-chain.

TEAM

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Problem

Document Forgery and Bureaucracy

Forgery Risks

The growing market of fake diplomas and certificates reduces employers' trust in qualifications.

Slow Verification

Manual verification through official requests to universities takes 5 to 14 business days.

Personal Data Leaks

Sending documents by email violates the requirements of the Law of the Republic of Kazakhstan “On Personal Data”.

INNOVATIVE MODEL

CertiChain Solution: Off-Chain + On-Chain

Solution

Why Is Blockchain Necessary?

- **Immutability:** A SHA-256 hash record cannot be forged or deleted retroactively.
- **Zero personal data disclosure:** Only a 64-character cryptographic signature is stored on-chain.
- **Instant validation:** An employer compares the file hash in 1 second without a centralized server.

Data Separation

ON-CHAIN (In the Blockchain)
File hash, university ID, timestamp, revocation status.

OFF-CHAIN (Local/IPFS)
Original PDF document, graduate's full name, grades.

COMMERCIAL STRATEGY

Two-Sided Business Model (B2B)

Monetization

Universities (Issuance)

500 KZT

Fee for each diploma registered upon issuance by an accredited university.

Employers (Verification)

1 200 KZT

Fee for a one-time cryptographic verification of a candidate by the HR department.

HR Subscription

150 000 KZT

Monthly package for large corporations (up to 200 verifications per month).

Solution Architecture and Smart Contracts

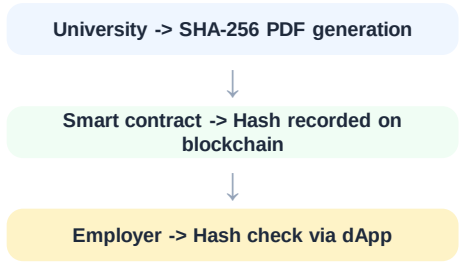
Platform: Polygon (Permissioned Node)

Low transaction fees ($0.002/transaction), high speed (2000 TPS) and block finality.

Smart contract functions:

- `issueCertificate(bytes32 docHash)`
- `verifyCertificate(bytes32 docHash)`
- `revokeCertificate(bytes32 docHash)`

Data Flow Diagram



Finance

Initial Capital (CAPEX & Initial OPEX)

REQUIRED INVESTMENT

18 500 000 KZT

Starting budget for 6 months until the project becomes self-sustaining.

Expense Category	Amount (KZT)	Share (%)
Development and Smart Contracts	9 000 000	48.6%
Marketing and B2B Sales	4 000 000	21.6%
Legal Support (AIFC/RK)	2 500 000	13.5%
Operating Reserve (Cash Reserve)	3 000 000	16.3%

Table 1. Allocation of initial capital. Source: team's own development.

Revenue and Cost Forecast (3 Years)

Indicator (in thousand KZT)	Year 1	Year 2	Year 3
Connected universities (pcs.)	5	15	35
Revenue from universities (registrations)	7 500	22 500	52 500
Revenue from employers (verifications)	9 600	36 000	90 000
Total Revenue	**17 100**	**58 500**	**142 500**
Operating expenses (OPEX)	19 200	32 000	58 000
Net Profit	**-2 100**	**26 500**	**84 500**

Table 2. Forecast of financial indicators for 2027-2029. Source: team's own financial model.

UNIT ECONOMICS

Analysis

Unit Economics and Break-Even Point

Unit Economics (1 Verification)

HR verification price: 1 200 KZT

Gas Fee (Polygon): ~2 KZT

Server / AWS / API: ~18 KZT

Margin: 98.3% (1 180 KZT)

BREAK-EVEN POINT

1 350

verifications per month

Break-even is planned for the 9th month of the startup's operation.

REGULATORY ANALYSIS

Legal Security and Regulatory Compliance

Legal

Law of the RK “On Personal Data”

The absence of full names and IINs in the public blockchain registry rules out violations of RK legislation.

Law of the RK “On Digital Assets”

The startup does not issue cryptocurrencies or conduct financial token exchange. The model is a utility SaaS.

AIFC Rules

Simulation nature of the project. Full compliance with the course's internal security requirements.

Interim Conclusions and Risks

Weak Assumptions

Readiness of Kazakhstan's conservative universities to integrate the API. Development of pilot regulations is required.

Plan for the Final Project

Development of a working MVP prototype on a local network, a Solidity smart contract, and an investment pitch.

CONCLUSION

Conclusion

Summary

- ✓ The CertiChain model solves the problem of diploma forgery with minimal fees and instant verification.
- ✓ Off-chain/on-chain hash storage fully protects personal data and complies with RK laws.
- ✓ The financial model shows net profit from Year 2 (26.5 million KZT) and high scaling potential.

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